AI AGENT EVALUATION AND OBSERVABILITY PLATFORM FOR RELIABLE RAG-BASED APPLICATIONS

V.S.B. ENGINEERING COLLEGE (AUTONOMOUS),
KARUR
DEPARTMENT OF COMPUTER AND COMMUNICATION ENGINEERING

GUIDE : [NAME & DESIGNATION]
TEAM MEMBERS :
SAKTHI MANOJ M
(922524118046)

UVANESWARN B
(922524118059)

ABSTRACT

- Large Language Models (LLMs), Retrieval-Augmented Generation (RAG) and tool-using AI agents are widely used for document question answering and task automation, but fluent responses do not guarantee factual accuracy, evidence grounding or correct tool execution. Failures can originate in irrelevant retrieval, unsupported claims, incorrect tool selection or inefficient execution, and manual inspection of final answers alone cannot reliably identify their causes. This project proposes an integrated platform to evaluate and observe RAG-based agents using a controlled test dataset. Each run is designed to capture the query, retrieved passages and similarity scores, tool calls, model response, latency, token usage and errors. The evaluation combines deterministic checks, LLM-as-a-Judge and a small human-validated sample to assess retrieval quality, answer relevance, faithfulness, tool selection and task completion. A dashboard will support trace inspection, failure analysis, model and prompt comparison, and repeatable regression testing. Actual performance results will be added after experiments are executed.

INTRODUCTION & OBJECTIVES

- Large Language Models can generate natural-language answers, while Retrieval-Augmented Generation (RAG) connects them to external documents and AI agents extend them through multi-step decisions and tool calls. Yet reliability depends on every stage: retrieving relevant evidence, selecting and using tools correctly, generating supported answers, and completing tasks within acceptable time and resource limits. This project aims to design an integrated platform for systematically evaluating and monitoring RAG-based agents. It will use a controlled dataset, capture detailed execution traces, measure retrieval and answer quality, assess tool selection and task completion, track latency, token usage and failures, compare model and prompt versions, and present findings through a dashboard to support diagnosis and regression testing.

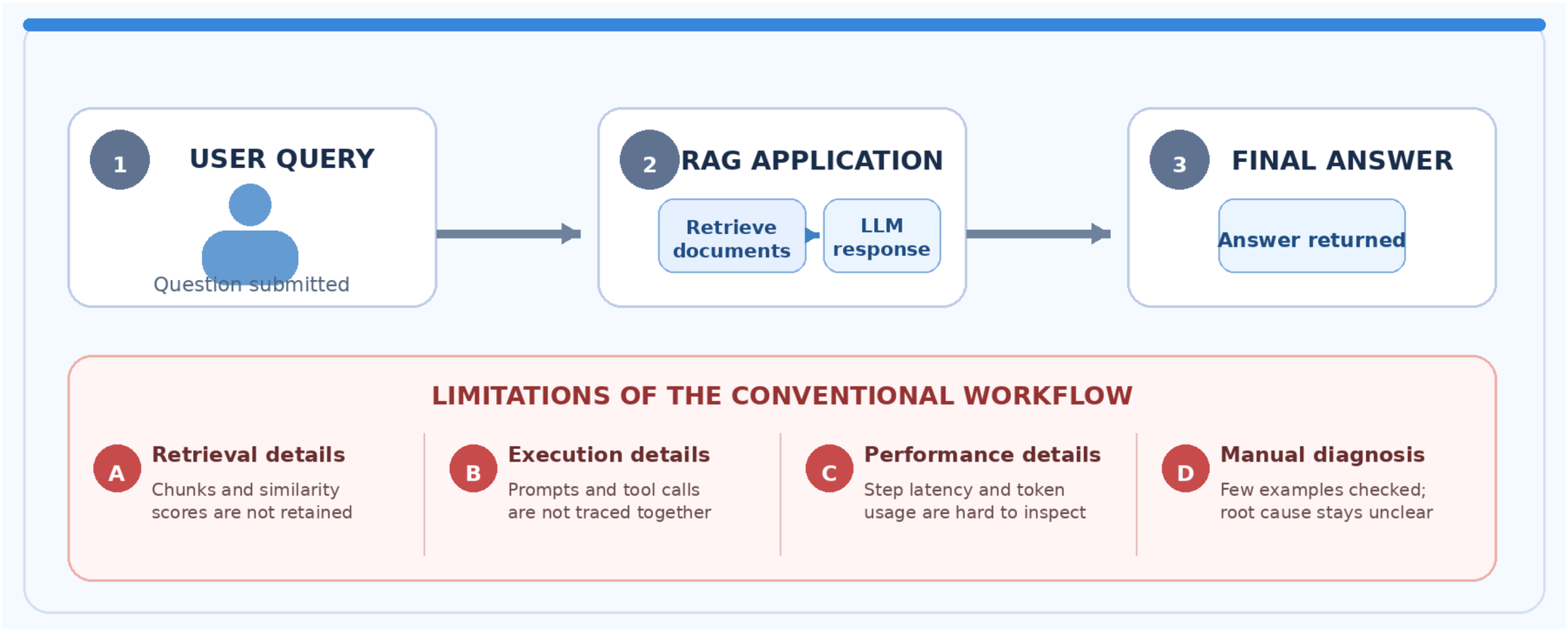
LITERATURE REVIEW (1/2)

S. NO.	TITLE	AUTHOR	YEAR	PROJECT LINK
1.	RAG for Knowledge-Intensive NLP Tasks	Lewis et al.	2020	Foundation for retrieval plus generation.
2.	RAGAs: Automated RAG Evaluation	Es et al.	2024	Supports faithfulness and relevance metrics.
3.	ARES: Automated Evaluation for RAG	Saad-Falcon et al.	2024	Informs human calibration of automated judges.

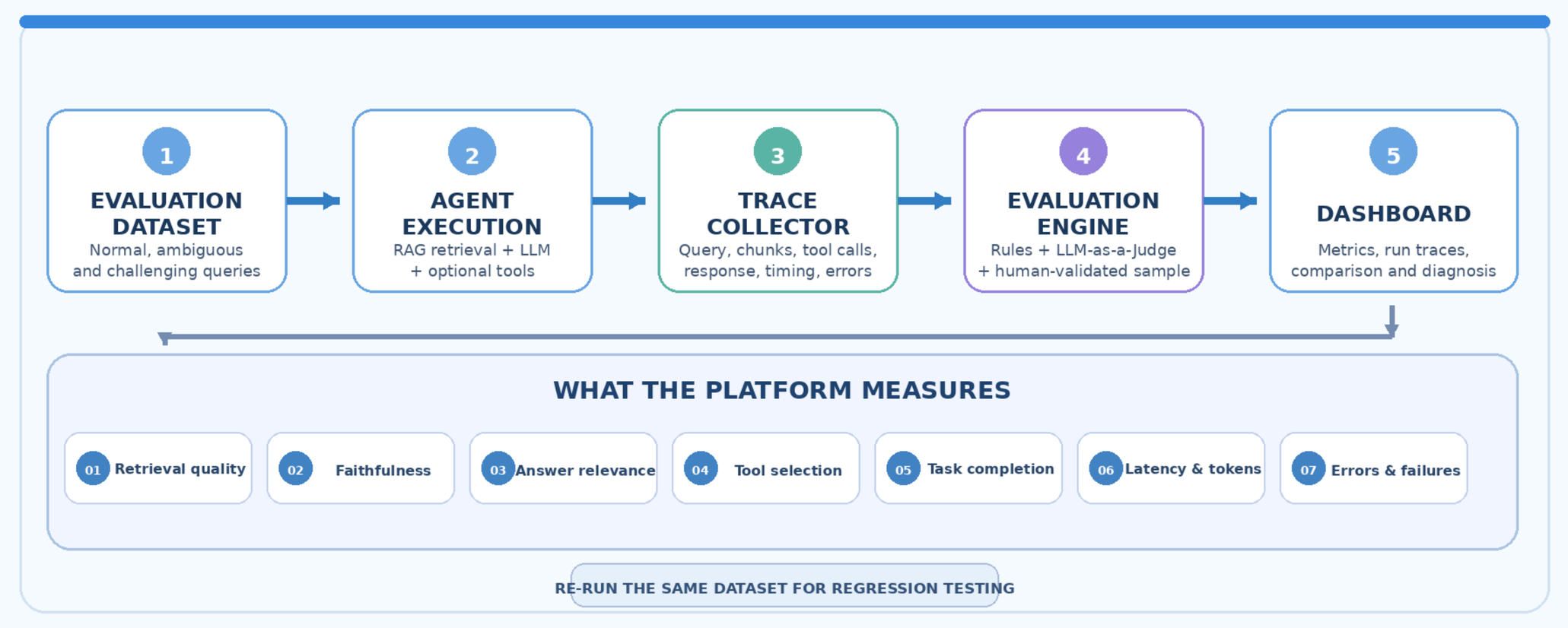
LITERATURE REVIEW (2/2)

S. NO.	TITLE	AUTHOR	YEAR	PROJECT LINK
4.	RAGChecker: Fine-Grained RAG Diagnosis	Ru et al.	2024	Separates retrieval and generation errors.
5.	AgentBench: Evaluating LLM Agents	Liu et al.	2024	Motivates multi-step agent and task metrics.

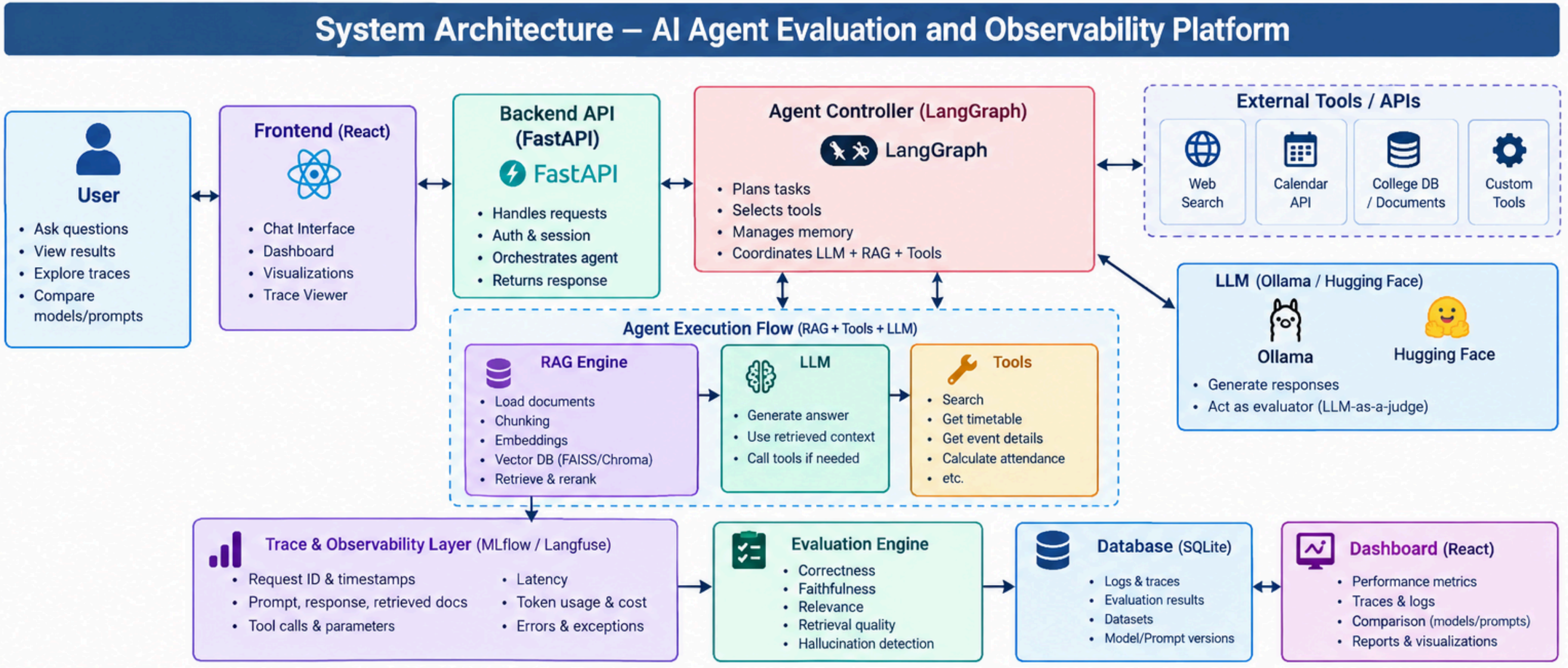
EXISTING SYSTEM



PROPOSED SYSTEM



SYSTEM BLOCK DIAGRAM



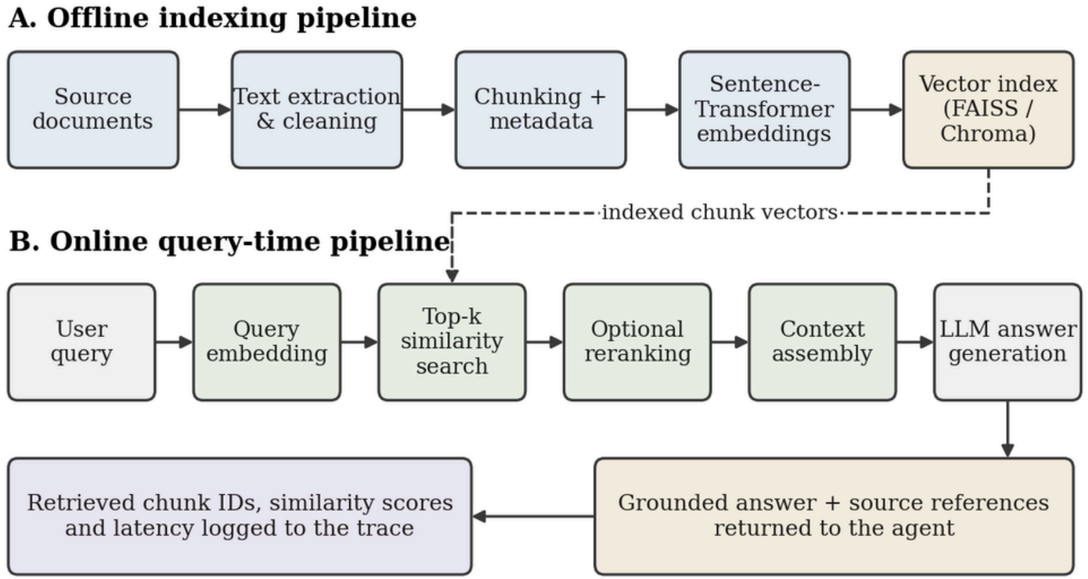
MODULE 1 – RAG & AGENT EXECUTION

- **Document preparation: load files, clean text, create chunks and retain document, section and page metadata.**

- **Embedding and retrieval: Sentence Transformers creates vectors; FAISS or Chroma retrieves Top-K chunks, with optional reranking.**

- **Agent execution: LangGraph coordinates query analysis, retrieval, optional tools and LLM generation through Ollama or Hugging Face.**

- **Output: a grounded answer with source references; model and retrieval settings are recorded for repeatable experiments.**



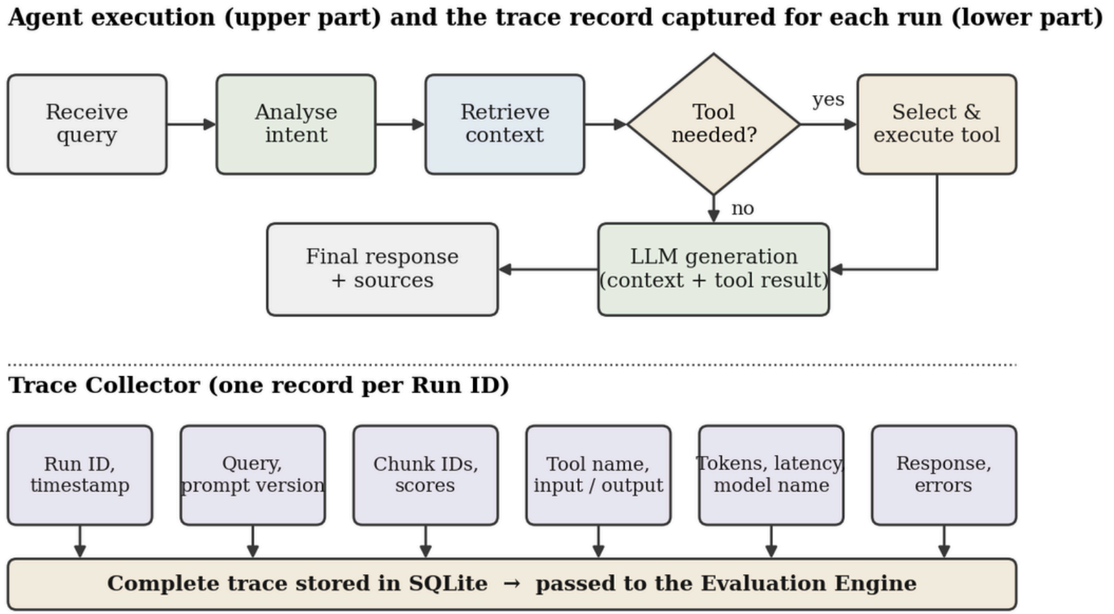
MODULE 2 – TRACE & EVALUATION

- **Trace collection: store a Run ID, timestamp, query, prompt/model version, retrieved chunk IDs and scores, tool inputs/outputs, latency, token usage and errors.**

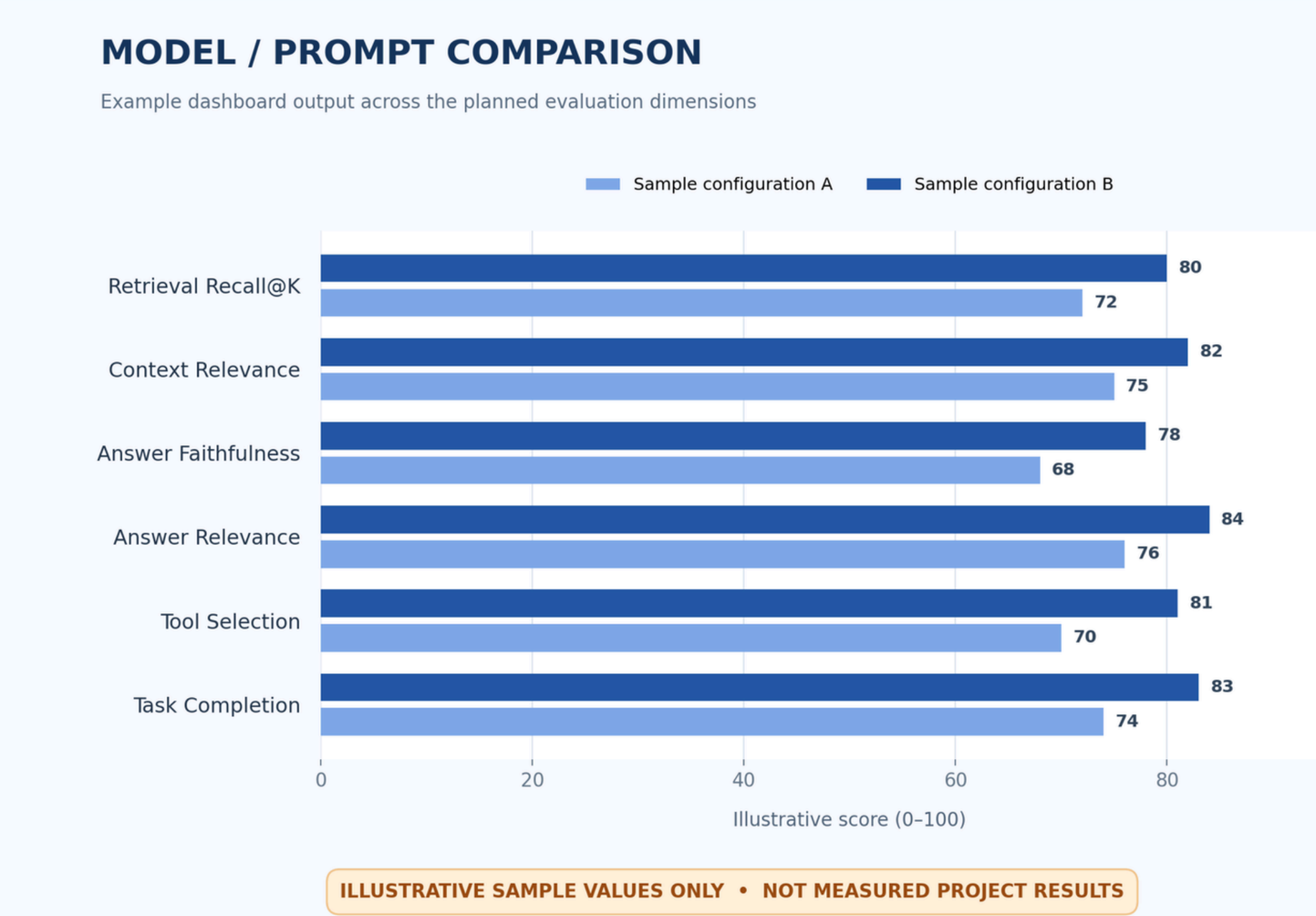
- **Evaluation engine: calculate Recall@K and tool/task checks when labels exist; use model-based scoring for faithfulness, relevance and answer quality; calibrate with human review.**

- **Dashboard: compare prompts and models, inspect per-step latency, explore traces and classify retrieval, tool, generation or infrastructure failures.**

- **Status: this review defines the design and test plan; measured scores remain pending until experiments are executed.**



RESULTS – EVALUATION DASHBOARD



CONCLUSION

- The proposed platform treats a RAG-based agent as a measurable workflow rather than a black box.
- It connects retrieval, tool interactions and LLM generation traces with quality, task and performance metrics.
- The dashboard is designed to support prompt/model comparison, failure diagnosis and repeatable regression testing.
- This review presents the architecture and evaluation plan; implementation results will be reported after experiments are completed.

FUTURE WORK

- **Expand the evaluation dataset and increase the human-validated reference cases.**
- **Add hybrid retrieval, reranking and finer retrieval/generation diagnostics.**
- **Explore multi-agent testing, automated prompt optimization and model routing.**
- **Integrate CI/CD regression evaluation, red-team testing, distributed tracing and production alerts.**
- **Run larger experiments across additional models and application domains.**

REFERENCES

- [1] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” NeurIPS, 2020.
- [2] S. Es et al., “RAGAs: Automated Evaluation of Retrieval Augmented Generation,” EACL System Demonstrations, 2024.
- [3] R. Saad-Falcon et al., “ARES: An Automated Evaluation Framework for RAG Systems,” NAACL-HLT, 2024.
- [4] Y. Ru et al., “RAGChecker: A Fine-grained Framework for Diagnosing RAG,” NeurIPS, 2024.
- [5] X. Liu et al., “AgentBench: Evaluating LLMs as Agents,” ICLR, 2024.

THANK YOU !!